INTRODUCTION REPORT

Air Passenger Compensation Claim Triage Service

ML System Design & AgentOps (UE24AM342AA5) | PES University | CSE (AI & ML)

1. Project Overview

This project proposes an AI-powered multi-agent service to analyze passenger compensation claims for flight delays, cancellations, denied boarding and downgrading. It will validate claim details, retrieve relevant DGCA Civil Aviation Requirements and a selected airline passenger charter, and generate evidence-backed entitlement recommendations with clause-level citations. Decisions requiring further evidence or human judgment will be escalated; the service will not authorize payments.

2. Team Responsibilities and Proposed Technologies

Aryaman DK | PES1UG24AM344

Backend, REST APIs, validation and asynchronous request processing

Tools: Python, SQL | FastAPI, Pydantic, Redis, PostgreSQL

Shristi Shukla | PES1UG24AM268

Multi-agent reasoning, claim classification, verification and escalation

Tools: Python | LangGraph, LangChain, LLM API

Spoorthy M | PES1UG24AM280

DGCA document processing, RAG retrieval and evaluation scenarios

Tools: Python | LangChain, ChromaDB, Sentence Transformers

Aditya Umashankar | PES1UG24AM335

Web interface, containerization, monitoring and load tests

Tools: JavaScript, HTML, CSS, Python | React, Docker, Grafana, Locust

3. Proposed Development Approach

The team will first confirm the DGCA and airline reference documents and agree on a shared claim schema and API contracts. Work will then proceed in parallel across backend, agent orchestration, retrieval and frontend/deployment. The modules will be integrated through FastAPI and tested using synthetic claim scenarios, including cases that require a second rules search or human escalation. Docker Compose will run the services locally, with tracing, monitoring, evaluation and load testing added before the demonstration.

4. Expected Outcome

A locally deployable web-based triage system that accepts passenger claims, produces verified recommendations supported by regulatory citations, requests missing evidence or escalates uncertain cases, and demonstrates core ML system design, AgentOps and LLMOps practices.